

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

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Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

Objective

- To implement a **Decision Tree classifier**.
- To extend the Decision Tree into a **Random Forest ensemble model**.
- To study the impact of hyperparameters on overfitting and generalization.
- To **select optimal hyperparameters using 5-Fold Cross-Validation**.
- To compare single-tree and ensemble-tree models.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset (loaded from `sklearn.datasets`)

- Total samples: 569 (357 Benign, 212 Malignant)
- Features: 30 numerical attributes
- Target classes: Malignant (0) and Benign (1)
- Train/Test split: 80%/20% (455 train, 114 test)

Theory

Decision Tree Classifier

A Decision Tree is a supervised learning model that recursively splits the feature space using impurity measures (Gini or Entropy) to form decision rules. It is prone to high variance and overfitting at large depths.

Random Forest Classifier

Random Forest is an ensemble of decision trees trained on bootstrapped samples with random feature selection. Majority voting aggregates predictions, reducing variance and improving generalization.

Steps for Implementation

1. Load dataset and verify class distribution.
2. Perform EDA: class distribution plot and correlation heatmap.
3. 80–20 train/test split with stratification.
4. Decision Tree: 5-Fold CV over criterion $\times$ max_depth $\times$ min_samples_split.
5. Select best DT hyperparameters and retrain on full train set.
6. Random Forest: 5-Fold CV over n_estimators $\times$ max_depth $\times$ max_features.
7. Select best RF hyperparameters and retrain.
8. Evaluate both on the held-out test set.

Hyperparameter Tuning Results

Decision Tree Cross-Fold Results

Table 1: Decision Tree – 5-Fold CV Results

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
Gini	3	92.53	0.9400
Gini	5	91.87	0.9335
Entropy	5	92.97	0.9439
Entropy	None	92.97	0.9440
Gini	7	91.87	0.9333

Best DT: criterion=entropy, max_depth=5, min_samples_split=2

Random Forest Cross-Fold Results

Table 2: Random Forest – 5-Fold CV Results

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
50	5	sqrt	95.60	0.9647
100	10	sqrt	96.26	0.9699
100	None	sqrt	96.26	0.9699
200	None	sqrt	96.26	0.9700
100	None	log2	96.48	0.9718

Best RF: n_estimators=100, max_depth=None, max_features=log2

5-Fold Cross-Validation Performance Comparison

Table 3: 5-Fold CV Accuracy per Fold (Best Models)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	92.98%	91.23%	97.37%	94.74%	92.92%	93.85%
Random Forest	96.49%	95.61%	95.61%	95.61%	96.46%	95.96%

Final Test-Set Evaluation

Table 4: Test-Set Performance (114 samples)

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Decision Tree	92.98%	0.9706	0.9167	0.9429	0.9435
Random Forest	95.61%	0.9589	0.9722	0.9655	0.9924

Observation Questions

- **How does tree depth affect overfitting?** Shallow trees (depth=3–5) generalise better. Unconstrained depth (=None) fits training data perfectly but may overfit; here, with only 30 features, the effect is mild because the data is not noisy.
- **Which hyperparameter had the greatest impact?** The choice of `max_features` in Random Forest (`log2` vs. `sqrt`) marginally improved accuracy. For Decision Tree, `criterion=entropy` with `max_depth=5` gave the best balance.
- **How does Random Forest improve generalisation?** By training 100 trees on different bootstrap samples with random feature subsets, variance is reduced via averaging, leading to a 2.6% accuracy gain over a single tree.
- **Did ensemble learning always improve performance?** Yes – Random Forest consistently outperformed the single Decision Tree across all 5 folds and on the final test set.

Conclusion

Decision Tree (entropy, depth=5) achieved 92.98% test accuracy; Random Forest (100 trees, log2 features) achieved 95.61% with an AUC of 0.9924 – a substantial improvement demonstrating the power of ensemble averaging for variance reduction.

References

- Scikit-learn: Decision Trees
- Scikit-learn: Random Forest
- UCI Dataset